

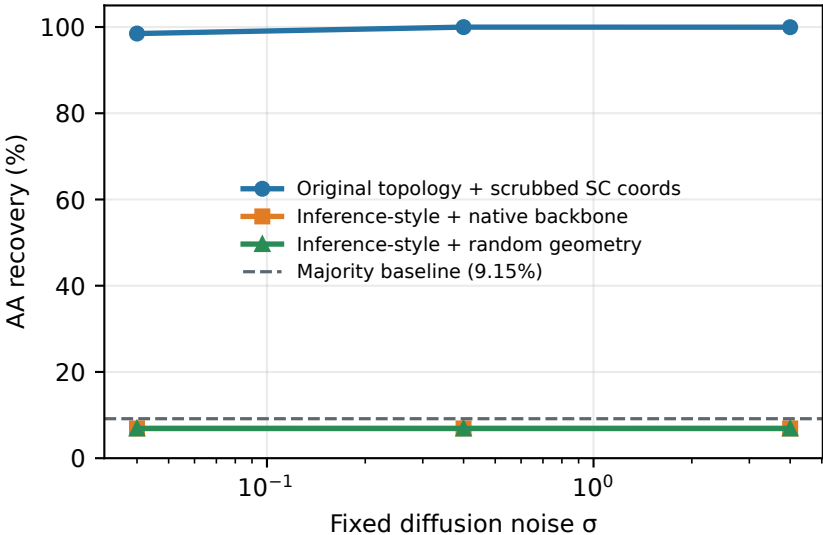
Finding The original 98.5–100.0% accuracy is driven by native atom topology/reference metadata. With inference-style inputs, accuracy is 6.9% and native geometry is no better than random geometry.

1. Input differences

Feature	Original training / validation	Inference-style validation
Sequence token	All design residues masked as xpb	All design residues masked as xpb
Coordinates	Noised native structure	Noised native N/Cα/C/O backbone*
Atom rows	Native full-atom rows retained	Exactly N/Cα/C/O; side chains removed
Reference metadata	Native AA-specific topology, names, elements, charges and ref positions	Uniform canonical backbone metadata
Sequence-derived features	MSA/profile masked, but native atom metadata remains	MSA/profile/template/hotspot zeroed
Native labels	aa_clean present for the training loss	Stored outside model input; metrics only

*Native backbone coordinates are used only so predictions can be compared with a known sequence; free monomer inference starts from Gaussian coordinates.

2. Validation result



Accuracy by noise level

Condition	$\sigma = 0.04$	$\sigma = 0.4$	$\sigma = 4$
Original topology + scrubbed SC coords	98.48%	99.97%	99.96%
Inference-style + native backbone	6.90%	6.90%	6.90%
Inference-style + random geometry	6.90%	6.90%	6.90%

Interpretation

Keeping native atom metadata preserves near-perfect prediction even after side-chain coordinates are scrubbed. Removing that metadata while retaining the native backbone collapses performance to the random-geometry control and below the 9.15% residue-prior baseline. The reported 90%+ result therefore does not measure backbone-to-sequence prediction.